

Day 2 Solution: Concatenation of Array

Problem Explanation:

We are given a array or in python we call it a list.

The goal is to concatenate/combine/add the list with itself, meaning if the input array is [1, 2, 3], the output should be [1, 2, 3, 1, 2, 3].

Methods To Solve This Problem:

Here, we discuss 5 different methods to concatenate an array with itself in Python.

1. Using Loop and Append Method
2. Using Loop and Changing Element Value
3. Using Extend Method
4. Using + Operator
5. Using * Operator

Method 1: Using Loop and Append Method

In this method, we use a loop to add each element from the original list to a new list twice.

1. Start with an empty list called result.
2. Loop through the original list and append each element to result.
3. Loop through the original list again and append each element to result.
4. Return the result list.

Code:

```
class Solution:
    def getConcatenation(self, nums: List[int]) -> List[int]:
        result = []
        for num in nums:
            result.append(num) # Add each element from nums
        for num in nums:
            result.append(num) # Add each element again
        return result
```

Method 2. Using Loop and Changing Element Value

In this method, we create a new list of the required size and use a loop to set each element.

1. Determine the length of the original list.
2. Create a new list called result with twice the length of the original list.
3. Use a loop to set each element in the new list:
 - First, copy elements from the original list.
 - Then, copy elements from the original list again, starting at the next index.
4. Return the result list.

Code:

```
class Solution:
    def getConcatenation(self, nums: List[int]) -> List[int]:
        n = len(nums)
        result = [0] * (2 * n) # Create a new list of size 2n
        for i in range(n):
            result[i] = nums[i] # Copy the first part
            result[i + n] = nums[i] # Copy the second part
        return result
```

Method 3: Using Extend Method

In this method, we use the extend method to add elements from the original list to itself.

1. Use the extend method to add elements from the original list to itself.
2. Return the modified list.

Code:

```
class Solution:
    def getConcatenation(self, nums: List[int]) -> List[int]:
        nums.extend(nums) # Extend the list with itself
        return nums
```

Method 4: Using + Operator

In this method, we use the + operator to concatenate the list with itself.

1. Use the + operator to concatenate the list with itself.
2. Return the new list.

Code:

```
class Solution:
    def getConcatenation(self, nums: List[int]) -> List[int]:
        return nums + nums # Concatenate the list with itself
```

Method 5: Using * Operator

In this method, we use the * operator to create a new list with the original list repeated twice.

1. Use the * operator to repeat the list twice.
2. Return the new list.

Code:

```
class Solution:  
    def getConcatenation(self, nums: List[int]) -> List[int]:  
        return nums * 2 # Repeat the list twice
```

Complexity:

- Time complexity: $O(n)$. In all the methods, we iterate through the list `nums` once, performing constant-time operations for each element.
- Space complexity: $O(n)$, as they require additional space proportional to the size of the input list to hold the concatenated result.

Hope this helps!